Nkechi Umeadi

Embedded Systems Engineer

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SUMMARY

Embedded engineer working on solar-powered metering and monitoring devices for off-grid communities. Four years in the field; I enjoy robust, low-power designs that survive heat, dust and unreliable networks. I am a hands-on problem solver who is not afraid of travelling to remote sites to diagnose devices. I enjoy working with limited hardware, patchy networks and practical constraints, and I document what I learn for the next engineer.

SKILLS

C, ESP32, STM32, MQTT, GSM modules, Low-power design, PCB debugging, Modbus, RS-485, Altium (reading schematics), Git, Python

EXPERIENCE

Embedded Systems Engineer, Sunstream Energy Devices 2021 – Present

- Wrote firmware for prepaid solar meters with GSM connectivity and local fallback when the network drops.
- Added watchdog and brown-out handling that cut field resets by 80%.
- Collaborated with hardware engineers on board revisions and field failure analysis.
- Documented a repeatable flashing and test procedure for the factory.
- Train field technicians to install and troubleshoot meters, and write simple guides in English and Igbo.

Electronics Intern, Eastern Power Systems 2020 (8 months)

- Assisted with prototype testing and soldering.
- Logged sensor data with Arduino based loggers.
- Wrote firmware for a temperature logger used in a cold storage pilot.
- Presented a final project that won a departmental prize.
- Validated new component suppliers by building small test fixtures and comparing failure rates.

Student Researcher, University of Enugu Technology 2019 – 2020

- Built a prototype energy monitor for a final year project.
- Tested sensors and calibrated readings against a reference meter.
- Wrote the thesis and presented results to a panel.
- Helped build a solar charge controller prototype and tested it outdoors in various weather conditions.
- Wrote weekly lab reports summarising test results.

EDUCATION

- BEng Electrical and Electronic Engineering, University of Enugu Technology (2015 – 2020)

LANGUAGES

English (native), Igbo (native)